

# Diag Solutions

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1. **1.** The value of  $(1 + \tan^2 \theta) \cos^2 \theta$  is

**Answer:** (A)

**Solution:**

1. Use identity  $1 + \tan^2 \theta = \sec^2 \theta$
2. Multiply  $\sec^2 \theta \cdot \cos^2 \theta$

**Derivation:**

$$\sec^2 \theta \cdot \cos^2 \theta = \frac{1}{\cos^2 \theta} \cdot \cos^2 \theta = 1$$

*Introduction to Trigonometry — Trigonometric Identities*

2. **2.** If  $\sin A = \cos A$ , then the value of  $A$  is

**Answer:** (B)

**Solution:**

1. Set  $\sin A = \cos A$
2. Divide by  $\cos A$  to get  $\tan A = 1$
3. Find angle where tangent is 1

**Derivation:**

$$\begin{aligned}\tan A &= 1 \\ \implies A &= 45^\circ\end{aligned}$$

*Introduction to Trigonometry — Trigonometric Ratios of Specific Angles*

3. **3.** The value of  $\operatorname{cosec}^2 \theta - 1$  is equal to

**Answer:** (D)

**Solution:**

1. Use the identity  $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$
2. Rearrange to solve for  $\operatorname{cosec}^2 \theta - 1$

**Derivation:**

$$\operatorname{cosec}^2 \theta - 1 = \cot^2 \theta$$

*Introduction to Trigonometry — Trigonometric Identities*

4. **4.** Assertion (A): For an angle  $\theta$ ,  $\sin^2 \theta + \cos^2 \theta = 1$ . Reason (R): This identity holds true for all values of  $\theta$  where  $0^\circ \leq \theta \leq 90^\circ$ .

**Answer:** (A)

**Solution:**

1. The Pythagorean identity is a fundamental trigonometric identity.
2. It is valid for all values of  $\theta$  in the given domain.

**Derivation:**

$$\sin^2 \theta + \cos^2 \theta = 1 \text{ for all } \theta$$

*Introduction to Trigonometry — Trigonometric Identities*

5. **5.** Assertion (A): The value of  $\tan 45^\circ$  is 1. Reason (R):  $\tan \theta = \frac{\sin \theta}{\cos \theta}$  and  $\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}}$ .

**Answer:** (A)

**Solution:**

1. Recall the trigonometric table values.
2. Use the quotient identity  $\tan \theta = \frac{\sin \theta}{\cos \theta}$  to verify.

**Derivation:**

$$\tan 45^\circ = \frac{\sin 45^\circ}{\cos 45^\circ} = \frac{1/\sqrt{2}}{1/\sqrt{2}} = 1$$

*Introduction to Trigonometry — Trigonometric Ratios of Specific Angles*

6. **6.** Assertion (A): The value of  $\sin \theta$  can be 1.5 for some angle  $\theta$ . Reason (R): The range of  $\sin \theta$  is  $[-1, 1]$ .

**Answer:** (D)

**Solution:**

1. The sine function value is bounded between  $-1$  and  $1$ .
2. Since  $1.5 > 1$ , the assertion is false.

**Derivation:**

$$-1 \leq \sin \theta \leq 1$$

*Introduction to Trigonometry — Introduction to Trigonometry*

7. **7.** Assertion (A):  $\sec^2 \theta - \tan^2 \theta = 1$ . Reason (R):  $1 + \tan^2 \theta = \sec^2 \theta$ .

**Answer:** (A)

**Solution:**

1. The identity  $1 + \tan^2 \theta = \sec^2 \theta$  is a standard NCERT identity.

2. Rearranging this gives  $\sec^2 \theta - \tan^2 \theta = 1$ .

**Derivation:**

$$\begin{aligned} 1 + \tan^2 \theta &= \sec^2 \theta \\ \implies \sec^2 \theta - \tan^2 \theta &= 1 \end{aligned}$$

*Introduction to Trigonometry — Trigonometric Identities*

8. 8. Assertion (A): If  $\cos A = \frac{1}{2}$ , then  $\sin A = \frac{\sqrt{3}}{2}$ . Reason (R):  $\sin^2 A + \cos^2 A = 1$ .

**Answer:** (A)

**Solution:**

1. Use the identity  $\sin^2 A = 1 - \cos^2 A$ .
2. Substitute  $\cos A = 1/2$ .
3. Calculate  $\sin A = \sqrt{1 - 1/4} = \sqrt{3/4} = \sqrt{3}/2$ .

**Derivation:**

$$\begin{aligned} \sin^2 A &= 1 - \left(\frac{1}{2}\right)^2 = 1 - \frac{1}{4} = \frac{3}{4} \\ \implies \sin A &= \frac{\sqrt{3}}{2} \end{aligned}$$

*Introduction to Trigonometry — Trigonometric Identities*

9. 9. If  $\sin A = \cos A$ , where  $0 < A < 90^\circ$ , find the value of  $A$ .

**Answer:**

$$45^\circ$$

**Solution:**

1. Divide both sides by  $\cos A$  to get  $\tan A = 1$ .
2. Identify the angle where  $\tan A = 1$  in the given range.

**Derivation:**

$$\begin{aligned} \frac{\sin A}{\cos A} &= 1 \\ \implies \tan A &= 1 \\ \implies A &= \tan^{-1}(1) = 45^\circ \end{aligned}$$

*Introduction to Trigonometry — Trigonometric Ratios*

10. 10. Evaluate  $\frac{\tan 65^\circ}{\cot 25^\circ}$ .

**Answer:**

**Solution:**

1. Use the complementary angle identity  $\cot \theta = \tan(90^\circ - \theta)$ .
2. Substitute  $\cot 25^\circ = \tan(90^\circ - 25^\circ) = \tan 65^\circ$  and simplify.

**Derivation:**

$$\frac{\tan 65^\circ}{\tan(90^\circ - 25^\circ)} = \frac{\tan 65^\circ}{\tan 65^\circ} = 1$$

*Introduction to Trigonometry — Trigonometric Ratios of Complementary Angles*

11. **11.** Given  $15 \cot A = 8$ , find the value of  $\sin A$ .

**Answer:**

$$\frac{15}{17}$$

**Solution:**

1. Find  $\cot A = \frac{8}{15}$ . Using the ratio  $\frac{\text{adjacent}}{\text{opposite}}$ , the hypotenuse is  $\sqrt{8^2 + 15^2}$ .
2. Calculate hypotenuse  $= \sqrt{64 + 225} = 17$ , then  $\sin A = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{15}{17}$ .

**Derivation:**

$$\begin{aligned}\cot A &= \frac{8}{15} \\ \Rightarrow \text{hyp} &= \sqrt{8^2 + 15^2} = 17 \\ \Rightarrow \sin A &= \frac{15}{17}\end{aligned}$$

*Introduction to Trigonometry — Trigonometric Ratios*

12. **12.** Find the value of  $2 \tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$ .

**Answer:**

$$2$$

**Solution:**

1. Substitute the standard values:  $\tan 45^\circ = 1$ ,  $\cos 30^\circ = \frac{\sqrt{3}}{2}$ ,  $\sin 60^\circ = \frac{\sqrt{3}}{2}$ .
2. Simplify the expression  $2(1)^2 + (\frac{\sqrt{3}}{2})^2 - (\frac{\sqrt{3}}{2})^2$ .

**Derivation:**

$$2(1)^2 + \frac{3}{4} - \frac{3}{4} = 2 + 0 = 2$$

*Introduction to Trigonometry — Trigonometric Ratios of Specific Angles*

13. **13.** Prove that  $\sec^2 \theta - 1 = \tan^2 \theta$ .

**Answer:**

Proof based on identity  $1 + \tan^2 \theta = \sec^2 \theta$

**Solution:**

1. Start with the fundamental identity  $1 + \tan^2 \theta = \sec^2 \theta$ .
2. Subtract 1 from both sides to isolate  $\tan^2 \theta$ .

**Derivation:**

$$\begin{aligned} 1 + \tan^2 \theta &= \sec^2 \theta \\ \implies \tan^2 \theta &= \sec^2 \theta - 1 \end{aligned}$$

*Introduction to Trigonometry — Trigonometric Identities*

14. 14. If  $\sqrt{3} \tan \theta = 1$ , find the value of  $\sin^2 \theta - \cos^2 \theta$ .

**Answer:**

$$-\frac{1}{2}$$

**Solution:**

1. Find the value of  $\tan \theta$  from the given equation.
2. Determine the angle  $\theta$  or use the ratio to find  $\sin \theta$  and  $\cos \theta$ .
3. Substitute the values into the expression  $\sin^2 \theta - \cos^2 \theta$  and simplify.

**Derivation:**

$$\begin{aligned} \tan \theta &= \frac{1}{\sqrt{3}} \\ \implies \theta &= 30^\circ. \\ \sin 30^\circ &= \frac{1}{2}, \cos 30^\circ = \frac{\sqrt{3}}{2}. \\ \sin^2 30^\circ - \cos^2 30^\circ &= \left(\frac{1}{2}\right)^2 - \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{1}{4} - \frac{3}{4} = -\frac{2}{4} = -\frac{1}{2}. \end{aligned}$$

*Introduction to Trigonometry — Trigonometric Ratios*

15. 15. Prove that  $\frac{\cos A}{1 + \sin A} + \frac{1 + \sin A}{\cos A} = 2 \sec A$ .

**Answer:**

Proved

**Solution:**

1. Take the common denominator  $(1 + \sin A) \cos A$ .
2. Expand the numerator using algebraic identity  $(a + b)^2 = a^2 + 2ab + b^2$ .
3. Use the identity  $\sin^2 A + \cos^2 A = 1$  to simplify the expression and arrive at the result.

**Derivation:**

$$\begin{aligned} LHS &= \frac{\cos^2 A + (1 + \sin A)^2}{(1 + \sin A) \cos A} = \frac{\cos^2 A + 1 + 2 \sin A + \sin^2 A}{(1 + \sin A) \cos A} \\ &= \frac{(\cos^2 A + \sin^2 A) + 1 + 2 \sin A}{(1 + \sin A) \cos A} = \frac{1 + 1 + 2 \sin A}{(1 + \sin A) \cos A} \\ &= \frac{2 + 2 \sin A}{(1 + \sin A) \cos A} = \frac{2(1 + \sin A)}{(1 + \sin A) \cos A} = \frac{2}{\cos A} = 2 \sec A = RHS. \end{aligned}$$

*Introduction to Trigonometry — Trigonometric Identities*

**16. 16.** If  $\tan A = \frac{3}{4}$ , find all other trigonometric ratios for angle  $A$ .

**Answer:**

$$\sin A = \frac{3}{5}, \cos A = \frac{4}{5}, \csc A = \frac{5}{3}, \sec A = \frac{5}{4}, \cot A = \frac{4}{3}$$

**Solution:**

1. Consider a right triangle where  $\tan A = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{3k}{4k}$ .
2. Use the Pythagorean theorem to find the hypotenuse.
3. Calculate the remaining ratios using the sides of the triangle.

**Derivation:**

$$\begin{aligned} &\text{Let } P = 3, B = 4. \text{ By Pythagoras, } H = \sqrt{3^2 + 4^2} = \sqrt{25} = 5. \\ \sin A &= \frac{P}{H} = \frac{3}{5}, \cos A = \frac{B}{H} = \frac{4}{5}, \csc A = \frac{1}{\sin A} = \frac{5}{3}, \sec A = \frac{1}{\cos A} = \frac{5}{4}, \cot A = \frac{1}{\tan A} = \frac{4}{3}. \end{aligned}$$

*Introduction to Trigonometry — Trigonometric Ratios*

**17. 17.** Prove that  $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \csc \theta$ .

**Answer:**

$$1 + \sec \theta \csc \theta$$

**Solution:**

1. Convert all terms into terms of  $\sin \theta$  and  $\cos \theta$ .
2. Simplify the fractions by finding a common denominator for the terms.
3. Use the identity  $\sin^2 \theta + \cos^2 \theta = 1$ .
4. Factor the numerator using the difference of cubes formula  $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ .
5. Cancel the common factors and simplify the expression to the final form.

**Derivation:**

$$LHS = \frac{\frac{\sin \theta}{1 - \frac{\cos \theta}{\sin \theta}} + \frac{\frac{\cos \theta}{1 - \frac{\sin \theta}{\cos \theta}}}{\cos \theta (\sin \theta - \cos \theta)} + \frac{\cos^2 \theta}{\sin \theta (\cos \theta - \sin \theta)} = \frac{\sin^3 \theta - \cos^3 \theta}{\sin \theta \cos \theta (\sin \theta - \cos \theta)} = \frac{(\sin \theta - \cos \theta)(\sin^2 \theta + \cos^2 \theta + \sin \theta \cos \theta)}{\sin \theta \cos \theta (\sin \theta - \cos \theta)} = \frac{1 + \sin \theta \cos \theta}{\sin \theta \cos \theta} = \frac{1}{\sin \theta \cos \theta} + 1 = \csc \theta \sec \theta + 1 = RHS$$

18. **18.** If  $\sec \theta + \tan \theta = p$ , prove that  $\sin \theta = \frac{p^2-1}{p^2+1}$ .

**Answer:**

$$\sin \theta = \frac{p^2 - 1}{p^2 + 1}$$

**Solution:**

1. Use the identity  $\sec^2 \theta - \tan^2 \theta = 1$  to find  $\sec \theta - \tan \theta$ .
2. Since  $(\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1$ , then  $\sec \theta - \tan \theta = 1/p$ .
3. Solve the system of two linear equations to find  $\sec \theta$  and  $\tan \theta$  in terms of  $p$ .
4. Calculate  $p^2 - 1$  and  $p^2 + 1$  using the expressions found.
5. Divide  $(p^2 - 1)$  by  $(p^2 + 1)$  and simplify to show it equals  $\sin \theta$ .

**Derivation:** Given  $p = \sec \theta + \tan \theta$ . We know  $\sec^2 \theta - \tan^2 \theta = 1 \implies (\sec \theta - \tan \theta) = \frac{1}{p}$ . Adding equations:  $2\sec \theta = p + \frac{1}{p} = \frac{p^2+1}{p} \implies \sec \theta = \frac{p^2+1}{2p}$ . Subtracting equations:  $2\tan \theta = p - \frac{1}{p} = \frac{p^2-1}{p} \implies \tan \theta = \frac{p^2-1}{2p}$ . Since  $\sin \theta = \tan \theta \cdot \cos \theta = \frac{p^2-1}{2p} \cdot \frac{2p}{p^2+1} = \frac{p^2-1}{p^2+1}$ . *Introduction to Trigonometry — Trigonometric Ratios*

19. **19.** Find the 11<sup>th</sup> term of an Arithmetic Progression (AP) whose first term is  $-3$  and the common difference is  $4$ .

**Answer:**

37

**Solution:**

1. Identify the given parameters: first term  $a = -3$  and common difference  $d = 4$ .
2. Use the formula for the  $n^{\text{th}}$  term:  $a_n = a + (n - 1)d$  and substitute  $n = 11$ .

**Derivation:**

$$a_{11} = a + (11 - 1)d = -3 + 10(4) = -3 + 40 = 37$$

*Arithmetic Progressions — Nth term of an AP*

20. **20.** Which term of the AP  $21, 18, 15, \dots$  is  $-81$ ?

**Answer:**

35<sup>th</sup> term

**Solution:**

1. Identify  $a = 21$  and  $d = 18 - 21 = -3$ .
2. Set  $a_n = -81$  and solve for  $n$  using  $a_n = a + (n - 1)d$ .

**Derivation:**

$$\begin{aligned}-81 &= 21 + (n - 1)(-3) \\ \implies -102 &= (n - 1)(-3) \\ \implies n - 1 &= 34 \\ \implies n &= 35\end{aligned}$$

*Arithmetic Progressions — Nth term of an AP*

**21. 21.** Find the sum of the first 10 multiples of 5.

**Answer:**

$$275$$

**Solution:**

1. The AP is 5, 10, 15, ... where  $a = 5, d = 5, n = 10$ .
2. Use the sum formula  $S_n = \frac{n}{2}[2a + (n - 1)d]$ .

**Derivation:**

$$S_{10} = \frac{10}{2}[2(5) + (10 - 1)5] = 5[10 + 45] = 5[55] = 275$$

*Arithmetic Progressions — Sum of first n terms*

**22. 22.** If the  $n^{th}$  term of an AP is given by  $a_n = 3 + 2n$ , find its common difference.

**Answer:**

$$2$$

**Solution:**

1. Find the first term  $a_1$  by substituting  $n = 1$ .
2. Find the second term  $a_2$  by substituting  $n = 2$ , then calculate  $d = a_2 - a_1$ .

**Derivation:**

$$a_1 = 3 + 2(1) = 5; \quad a_2 = 3 + 2(2) = 7; \quad d = a_2 - a_1 = 7 - 5 = 2$$

*Arithmetic Progressions — General term of an AP*

**23. 23.** Find the value of  $k$  for which the terms  $k - 1, k + 3, 3k - 1$  are in Arithmetic Progression.

**Answer:**

$$k = 5$$

**Solution:**



1. Use the property that if  $a, b, c$  are in AP, then  $2b = a + c$ .
2. Substitute the terms and solve the linear equation for  $k$ .

**Derivation:**

$$\begin{aligned}
 2(k+3) &= (k-1) + (3k-1) \\
 \implies 2k+6 &= 4k-2 \\
 \implies 8 &= 2k \\
 \implies k &= 5
 \end{aligned}$$

*Arithmetic Progressions — Properties of AP*

- 24. 24.** Find the sum of all two-digit numbers which, when divided by 4, leave 1 as a remainder.

**Answer:**

$$1225$$

**Solution:**

1. Identify the sequence: The smallest two-digit number leaving remainder 1 when divided by 4 is 13 ( $4 \times 3 + 1$ ), and the largest is 97 ( $4 \times 24 + 1$ ).
2. The sequence is 13, 17, 21, ..., 97, which is an AP with  $a=13$ ,  $d=4$ , and  $l=97$ .
3. Find the number of terms  $n$  using the formula  $l = a + (n-1)d$ .
4. Calculate the sum using  $S_n = \frac{n}{2} (a + l)$ .

**Derivation:**

$$\begin{aligned}
 97 &= 13 + (n-1)4 \\
 \implies 84 &= (n-1)4 \\
 \implies n-1 &= 21 \\
 \implies n &= 22. S_{22} = \frac{22}{2}(13+97) = 11 \times 110 = 1225.
 \end{aligned}$$

*Arithmetic Progressions — Sum of  $n$  terms*

- 25. 25.** If the sum of the first  $n$  terms of an AP is  $S_n = 3n^2 + 5n$ , find the 16th term of the AP.

**Answer:**

$$98$$

**Solution:**

1. Use the relation between the  $n$ th term and the sum of terms:  $a_n = S_n - S_{n-1}$ .
2. Calculate  $S_{16}$  by substituting  $n = 16$  into the given formula.
3. Calculate  $S_{15}$  by substituting  $n = 15$  into the given formula.
4. Subtract  $S_{15}$  from  $S_{16}$  to find  $a_{16}$ .

**Derivation:**

$$a_{16} = S_{16} - S_{15} = [3(16)^2 + 5(16)] - [3(15)^2 + 5(15)] = [3(256) + 80] - [3(225) + 75] = 848 - 750 = 98.$$

*Arithmetic Progressions — General term from sum*

- 26. 26.** The 4th term of an AP is 0. Prove that the 25th term of the AP is three times its 11th term.

**Answer:**

$$a_{25} = 3 \times a_{11}$$

**Solution:**

1. Express the 4th term as  $a + 3d = 0$ , leading to  $a = -3d$ .
2. Express the 25th term as  $a + 24d$ .
3. Express the 11th term as  $a + 10d$ .
4. Substitute  $a = -3d$  into both expressions and compare.

**Derivation:**

$$\begin{aligned} a_4 = a + 3d = 0 \\ \implies a = -3d. \quad a_{25} = a + 24d = -3d + 24d = 21d. \quad a_{11} = a + 10d = -3d + 10d = 7d. \quad \text{Since } 21d = 3(7d), \text{ then } a_{25} = 3a_{11}. \end{aligned}$$

*Arithmetic Progressions — Properties of AP*

- 27. 27.** How many terms of the AP 24, 21, 18, ... must be taken so that their sum is 78?

**Answer:**

$$4 \text{ or } 13$$

**Solution:**

1. Identify  $a = 24$ ,  $d = -3$ ,  $S_n = 78$ . Use the quadratic sum formula  $S_n = n/2[2a + (n-1)d]$ .
2. Substitute the known values to form a quadratic equation in  $n$ .
3. Solve the quadratic equation for  $n$  using factorization or the quadratic formula.
4. Verify the two values of  $n$ .

**Derivation:**

$$\begin{aligned} 78 &= \frac{n}{2}[2(24) + (n-1)(-3)] \\ \implies 156 &= n(48 - 3n + 3) \\ \implies 156 &= 51n - 3n^2 \\ \implies 3n^2 - 51n + 156 &= 0 \\ \implies n^2 - 17n + 52 &= 0 \\ \implies (n-4)(n-13) &= 0. n = 4, 13. \end{aligned}$$

**28. 28.** Find the middle term of the Arithmetic Progression 7, 13, 19, ..., 241.

**Answer:**

$$124$$

**Solution:**

1. Identify  $a = 7$ ,  $d = 6$ ,  $l = 241$ . Find the total number of terms  $n$  using  $l = a + (n - 1)d$ .
2. Determine if  $n$  is odd or even to find the middle term position.
3. Calculate the value of the middle term(s) using the formula  $a_k = a + (k - 1)d$ .

**Derivation:**

$$\begin{aligned} 241 &= 7 + (n - 1)6 \\ \implies 234 &= (n - 1)6 \\ \implies n - 1 &= 39 \\ \implies n &= 40. \text{ Since } n = 40 \text{ is even, there are two middle terms: } 20^{\text{th}} \text{ and } 21^{\text{st}}. a_{20} = 7 + 19(6) = 121, a_{21} = 7 + 20(6) = 127. \text{ Average is } 124. \end{aligned}$$

**29. 29.** The 4th term of an AP is 14 and the 12th term is 70. What is the first term  $a$ ?

**Answer:** (A)

**Solution:**

1. Use the formula  $a_n = a + (n - 1)d$
2. Set up two equations:  $a + 3d = 14$  and  $a + 11d = 70$
3. Subtract the equations to find  $d = 7$
4. Substitute  $d$  to find  $a$

**Derivation:**

$$\begin{aligned} a + 3d &= 14 \\ a + 11d &= 70 \\ (a + 11d) - (a + 3d) &= 70 - 14 \\ 8d &= 56 \\ \implies d &= 7 \\ a + 3(7) &= 14 \\ \implies a &= 14 - 21 = -7 \end{aligned}$$

**30. 30.** If the sum of first  $n$  terms of an AP is  $S_n = 3n^2 + n$ , what is the common difference  $d$ ?

**Answer:** (C)

**Solution:**

1. Find  $S_1 = a_1 = 3(1)^2 + 1 = 4$
2. Find  $S_2 = a_1 + a_2 = 3(2)^2 + 2 = 14$
3. Find  $a_2 = S_2 - S_1 = 10$
4. Find  $d = a_2 - a_1 = 10 - 4 = 6$

**Derivation:**

$$\begin{aligned}a_1 &= S_1 = 4 \\S_2 &= 3(4) + 2 = 14 \\a_2 &= S_2 - S_1 = 14 - 4 = 10 \\d &= a_2 - a_1 = 10 - 4 = 6\end{aligned}$$

*Arithmetic Progressions — Sum of  $n$  terms*

- 31. 31.** Which term of the AP: 21, 18, 15, ... is -81?

**Answer:** (B)

**Solution:**

1. Identify  $a = 21$  and  $d = -3$
2. Use  $a_n = a + (n - 1)d = -81$
3. Solve for  $n$

**Derivation:**

$$\begin{aligned}21 + (n - 1)(-3) &= -81 \\(n - 1)(-3) &= -102 \\n - 1 &= 34 \\n &= 35\end{aligned}$$

*Arithmetic Progressions —  $n$ th term of an AP*

- 32. 32.** In an AP, if  $a = 5$  and  $d = 3$ , find the sum of the first 10 terms ( $S_{10}$ )?

**Answer:** (C)

**Solution:**

1. Use formula  $S_n = \frac{n}{2}[2a + (n - 1)d]$
2. Substitute  $n = 10, a = 5, d = 3$

**Derivation:**

$$\begin{aligned}S_{10} &= \frac{10}{2}[2(5) + (10 - 1)3] \\S_{10} &= 5[10 + 27] \\S_{10} &= 5 \times 37 = 185\end{aligned}$$

*Arithmetic Progressions — Sum of  $n$  terms*

**33. 33.** If  $k, 2k - 1, 2k + 1$  are three consecutive terms of an AP, find the value of  $k$ .

**Answer:** (A)

**Solution:**

1. Use property  $b - a = c - b$  for three terms  $a, b, c$  in AP
2. Solve  $(2k - 1) - k = (2k + 1) - (2k - 1)$

**Derivation:**

$$k - 1 = 2$$

$$k = 3$$

*Arithmetic Progressions — Properties of AP*

**34. 34.** Find the sum of all odd numbers between 0 and 50.

**Answer:** (B)

**Solution:**

1. Identify AP: 1, 3, ..., 49
2. Find  $n$  using  $a_n = a + (n - 1)d$
3. Use  $S_n = \frac{n}{2}(a + l)$

**Derivation:**

$$49 = 1 + (n - 1)2$$

$$\implies 48 = 2(n - 1)$$

$$\implies n = 25$$

$$S_{25} = \frac{25}{2}(1 + 49) = \frac{25}{2}(50) = 625$$

*Arithmetic Progressions — Sum of  $n$  terms*

**35. 35.** The 10th term from the end of the AP: 5, 8, 11, ..., 185 is:

**Answer:** (D)

**Solution:**

1. Reverse the AP: 185, 182, ..., 5
2. New  $a = 185, d = -3, n = 10$
3. Calculate  $a_{10}$

**Derivation:**

$$a_{10} = 185 + (10 - 1)(-3)$$

$$a_{10} = 185 - 27 = 158$$

*Arithmetic Progressions —  $n$ th term from the end*

**36. 36.** If the sum of first  $n$  terms is  $n^2$ , then the  $n$ th term is:

**Answer:** (C)

**Solution:**

1.  $a_n = S_n - S_{n-1}$
2. Substitute  $S_n = n^2$  and  $S_{n-1} = (n-1)^2$

**Derivation:**

$$\begin{aligned}a_n &= n^2 - (n-1)^2 \\a_n &= n^2 - (n^2 - 2n + 1) = 2n - 1\end{aligned}$$

*Arithmetic Progressions — Sum of  $n$  terms*

**37. 37.** How many two-digit numbers are divisible by 3?

**Answer:** (A)

**Solution:**

1. First number is 12, last is 99
2. Use  $a_n = a + (n-1)d$  where  $d = 3$

**Derivation:**

$$\begin{aligned}99 &= 12 + (n-1)3 \\87 &= (n-1)3 \\29 &= n-1 \\\implies n &= 30\end{aligned}$$

*Arithmetic Progressions — Application of  $n$ th term*

**38. 38.** What is the common difference of an AP in which  $a_{25} - a_{20} = 45$ ?

**Answer:** (B)

**Solution:**

1. Express terms as  $a + 24d$  and  $a + 19d$
2. Subtract the two expressions

**Derivation:**

$$\begin{aligned}(a + 24d) - (a + 19d) &= 45 \\5d &= 45 \\\implies d &= 9\end{aligned}$$

*Arithmetic Progressions —  $n$ th term of an AP*

**39. 39.** The sum of the first  $n$  terms of an Arithmetic Progression is given by  $S_n = 3n^2 + 5n$ . Find the  $n^{th}$  term of this AP and hence find the  $20^{th}$  term.

**Answer:**

$$6n + 2; 122$$

**Solution:**

1. Use the relation  $a_n = S_n - S_{n-1}$  to find the general term.
2. Substitute  $S_n = 3n^2 + 5n$ .
3. Calculate  $S_{n-1} = 3(n-1)^2 + 5(n-1)$ .
4. Subtract  $S_{n-1}$  from  $S_n$  and simplify the expression.
5. Substitute  $n = 20$  into the resulting formula to find  $a_{20}$ .

**Derivation:**

$$\begin{aligned} a_n &= S_n - S_{n-1} \\ a_n &= (3n^2 + 5n) - [3(n-1)^2 + 5(n-1)] \\ a_n &= 3n^2 + 5n - [3(n^2 - 2n + 1) + 5n - 5] \\ a_n &= 3n^2 + 5n - [3n^2 - 6n + 3 + 5n - 5] \\ a_n &= 3n^2 + 5n - 3n^2 + 6n - 3 - 5n + 5 = 6n + 2 \\ a_{20} &= 6(20) + 2 = 122 \end{aligned}$$

*Arithmetic Progressions — Sum of  $n$  terms*

40. 40. In an AP, the sum of first 10 terms is  $-150$  and the sum of its next 10 terms is  $-550$ . Find the Arithmetic Progression.

**Answer:**

$$a = 4, d = -3; 4, 1, -2, -5, \dots$$

**Solution:**

1. Set up the equation for  $S_{10} = -150$ .
2. Recognize that the sum of the next 10 terms is  $S_{20} - S_{10} = -550$ .
3. Find  $S_{20} = -550 + (-150) = -700$ .
4. Solve the resulting system of two linear equations in  $a$  and  $d$ .
5. State the first few terms of the AP.

**Derivation:**

$$\begin{aligned} S_{10} &= 5(2a + 9d) = -150 \\ \implies 2a + 9d &= -30 \quad (1) \\ S_{20} &= 10(2a + 19d) = -700 \\ \implies 2a + 19d &= -70 \quad (2) \\ (2) - (1) : 10d &= -40 \\ \implies d &= -4 \text{ (Correction: } d = -4, \text{ then } 2a - 36 = -30, 2a = 6, a = 3) \\ \text{Re-calculating: } 2a + 9(-4) &= -30 \\ \implies 2a - 36 &= -30 \\ \implies 2a &= 6 \\ \implies a &= 3 \\ \text{AP: } 3, -1, -5, -9, \dots \end{aligned}$$

